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APPLIED RESEARCH

A Scalable Real-Time Multi-Camera Vehicle Tracking System for Urban Environments

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ABSTRACT Real-time Multi-Camera Vehicle Tracking (MCVT) systems are challenging to design and deploy due to the need for real-time processing at the edge, as well as issues such as high noise, low image resolution, and reidentification. With the significant advances of computer vision techniques and transformer-based models for image classification, new opportunities have emerged for MCVT. Moreover, the growth of urban populations and the increasing demand for sustainable mobility have intensified the need for scalable traffic monitoring solutions. Despite recent progress, several critical challenges remain unresolved, including unreliable re-identification under severe occlusions and viewpoint changes, limited temporal synchronisation between heterogeneous cameras, and the lack of architectures that operate efficiently on fog and edge infrastructures. In this paper, we propose a scalable, low-latency MCVT pipeline designed for heterogeneous CCTV networks deployed across the edge-fog continuum. The system integrates lightweight vehicle detection, transformer-based appearance embeddings, and topology- and travel-time-constrained cross-camera association within a modular deployment-aware architecture. Experimental validation on both public benchmarks and a real-world logistics hub demonstrates stable identity preservation, correctly identifying more than 80% of vehicle matches under deployment conditions while maintaining strict computational constraints. The system sustains real-time performance (20–29 FPS per stream) under CPU-based detection across 27 CCTV cameras and seven edge nodes, demonstrating scalable multi-camera tracking under production-level constraints.

INDEX TERMS Fog computing, multi-camera tracking, traffic monitoring, vehicle reidentification.

I. INTRODUCTION

Real-time multi-camera vehicle tracking (MCVT) in urban environments is a priority capability for smart city traffic management and large logistics hubs. It involves continuously detecting and following vehicles across a network of street camera feeds to understand traffic flow, monitor congestion, and improve safety. Unlike single-camera tracking, the multi-camera setting poses additional challenges: each camera may have a different viewpoint and resolution, video streams often suffer from noise (e.g., low light or weather effects), frequent occlusions occur in dense traffic, and associating vehicle identities across non-overlapping cameras is notoriously tricky. In a city-scale deployment,

a vehicle might leave the field of view of one camera and reappear minutes later on another camera, at a different angle and with a different background. The system must reliably determine if it is the exact vehicle despite changes in appearance and environment. This identity matching across disjoint cameras traditionally relies on visual reidentification cues or external markers; however, both can be unreliable under real-world conditions, such as varied angles, motion blur, and partial occlusion. As a result, robust MCVT remains an open problem, especially under real-time requirements.

The ability to scale up vehicle tracking to city-scale or large campus-scale is increasingly important. Modern smart city initiatives and large logistics hubs (e.g. distribution centres, ports, or wholesale markets) employ hundreds of cameras to monitor transportation activity. Traditional monolithic or cloud approaches (e.g. sending all video feeds to a central

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cloud server for processing) can face bandwidth bottlenecks and single points of failure. Instead, fog and edge computing paradigms are emerging, wherein computation is distributed across edge nodes closer to the cameras (e.g. on-site servers or smart camera hubs). Although it reduces latency and network load, it means the tracking algorithms must operate under resource-constrained conditions (limited GPU/CPU per node, limited memory). High-end deep learning models that excel in research settings may struggle in edge environments due to their demanding computational and memory requirements.

Recent advances in deep learning have indeed dramatically improved visual tracking accuracy. In single-camera Multi-Object Tracking (MOT), methods such as DeepSORT [1] combine a fast Kalman filter tracker with deep neural network appearance embedding to associate detections over time. Transformer-based trackers have taken this further: for instance, TrackFormer [2], TransTrack [3], and AttentionTrack [4] introduce transformer attention to detect and track multiple objects in video frames jointly. End-to-end transformer models, such as the Multiple-Object Tracking Transformer (MOTR) [5], treat object tracking as a direct set prediction problem, propagating learned track queries frame by frame. However, they come at a cost: the networks are large and require GPU acceleration to run in real-time. Deploying a MOTR-style model on dozens of camera feeds would likely overwhelm typical edge computing nodes. There is a growing interest in lightweight computer vision techniques that trade some accuracy for efficiency. For example, in static-camera scenarios, background subtraction and motion segmentation can serve as simpler vehicle detectors. BYTETrack [6] has demonstrated that improved data association strategies can substantially improve accuracy without altering the underlying detector, using low-score detection boxes to link previous ones. These approaches achieve strong results on academic benchmarks, particularly in handling occlusions and maintaining identities over extended video sequences.

In the domain of multi-camera tracking, most existing solutions follow a decoupled two-step approach: first, perform independent tracking on each camera, then link the tracklets belonging to the same object across cameras using appearance features or spatio-temporal reasoning. For example, the widely used CityFlow benchmark [7] defines a city-scale multi-camera vehicle tracking task where there is a detection and single-camera tracking system for each video, and a separate association algorithm matches vehicles between cameras. Appearance-based vehicle reidentification (ReID) is a key component in many methods. Deep learning ReID models are trained on datasets like VeRi-776 [8] or VRIC [9] to encode each vehicle's visual features into an embedding vector, ensuring that the exact vehicle produces similar embeddings across different cameras. License plate recognition and spatio-temporal constraints are often incorporated to improve robustness. For instance, if an OCR module can read a vehicle's license plate, it provides a nearly unique ID to corroborate the visual match, a technique used in

intelligent transport systems for decades. Reference [10] has even explored using vehicle trajectory prediction to aid cross-camera association, effectively estimating where and when a vehicle might reappear, to narrow down candidate matches.

The proposed approach is motivated by the gap between bleeding-edge accuracy and practical application in constrained environments. The system is a modular, scalable, and efficient MCVT system tailored for fog/edge computing. Instead of a monolithic system, the proposal is split into specialised modules: (1) a vehicle detection module using traditional computer vision techniques to identify vehicles in each camera frame quickly; (2) a feature extraction module that computes a compact appearance descriptor for each vehicle; and (3) a multi-source data association module that combines the appearance descriptors and spatio-temporal context to reidentify vehicles across cameras and reconstruct complete trajectories. By splitting the problem into these stages, each stage can be optimised or scaled independently. This modular design is inherently scalable, allowing more processing nodes to be added to handle additional camera streams or higher frame rates. The system can also degrade gracefully by skipping or simplifying specific modules if resources are constrained.

This work addresses real-time multi-camera vehicle tracking by proposing a deployment-aware architecture that integrates motion consistency, appearance learning, and physical plausibility constraints within an edge-fog computing framework. The contributions are threefold. First, we introduce a lightweight direction-aware centroid linking strategy for single-camera tracking that enforces multi-scale motion consistency without relying on deep tracking backbones. The proposed association logic combines spatial overlap gating with short-, medium-, and long-term directional constraints, enabling stable tracklet formation under edge computing limitations. Second, we design a topology- and travel-time-constrained cross-camera association framework that couples compact transformer-based appearance embeddings with physically grounded feasibility checks. This formulation reduces implausible cross-camera identity merges in non-overlapping CCTV networks. Third, we validate the complete system in a real-world deployment spanning 27 CCTV cameras and seven edge nodes, sustaining 20–29 FPS per stream under CPU-based detection. The results demonstrate scalable, low-latency multi-camera tracking under production-level constraints.

The paper is structured as follows. Section II reviews the related work on single- and multi-camera tracking, vehicle reidentification, and edge-fog deployment strategies. Section III presents the proposed multi-camera vehicle tracking architecture, detailing the detection, feature extraction, and trajectory reconstruction modules. Section IV reports the experimental evaluation, including detection accuracy, embedding quality, multi-camera reidentification performance, and a real-world deployment study. Finally, Section V concludes the paper and outlines directions for future work.

TABLE 1. Summary of acronyms and variables used throughout the paper.

Abbreviation	Full Form
AMQP	Advanced Message Queuing Protocol
BEV	Bird's Eye View
BEVFormer	Bird's Eye View Transformer
CCTV	Closed-Circuit Television
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DFS	Depth-First Search
DETR	Detection Transformer
DeepSORT	Deep Simple Online and Realtime Tracking
FPS	Frames Per Second
FOV	Field of View
GIAOTracker	Global and Instance-Aware Online Tracker
GPU	Graphics Processing Unit
IDF1	Identity F1-score
IDP	Identity Precision
IDR	Identity Recall
ImageNet	Large-Scale Image Recognition Dataset
IoU	Intersection over Union
KDE	Kernel Density Estimation
LoRA	Low-Rank Adaptation
LPR	License Plate Recognition
mAP	Mean Average Precision
MBFSR	Multi-Branch Feature Fusion Super-Resolution
MCT	Multicamera multi-object tracking
MCMOT	Multicamera multi-object tracking
MCVT	Multi-Camera Vehicle Tracking
MDFE-Net	Multiple Discriminative Features Enhancement Network
MOT	Multi-Object Tracking
MOTR	Multiple-Object Tracking Transformer
mPSO	mean Positive Sample Occupancy
NLP	Natural Language Processing
OCR	Optical Character Recognition
OC-SORT	Observation-Centric SORT
PETR	Position Embedding Transformation for multi-view 3D detection
R&D	Research and Development
RAG	Retrieval-Augmented Generation
ReID	Reidentification
ROI	Region of Interest
SISR	Single Image Super-Resolution
SORT	Simple Online and Realtime Tracking
SOTA	State of the Art
TransReID	Transformer-based Re-identification model
TransTrack	Transformer Tracking architecture
ViT	Vision Transformer
ViT-B/16	Vision Transformer Base with 16x16 patch size
VRIC	Vehicle Re-Identification in Context dataset
V2X	Vehicle-to-Everything communication
YOLO	You Only Look Once object detector
YOLO-JDE	YOLO Joint Detection and Embedding

Table 1 summarizes the main acronyms and variables used throughout the paper.

II. RELATED WORK

This section reviews the main research directions relevant to multi-camera vehicle tracking. It first summarizes advances in single-camera multi-object tracking that form the foundation for cross-camera association. Then, it discusses strategies for multi-camera tracking and data association across views, followed by recent developments in reidentification models that enable robust vehicle matching under real-world conditions. An overview of the main approaches, including

their assumptions, topological requirements, data association strategies, and evaluation paradigms, is summarized in Table 2.

A. MULTI-OBJECT TRACKING

MOT is traditionally formulated as a tracking-by-detection problem: objects are first detected in each frame, and then these detections are linked over time to form trajectories. Early approaches relied on simple appearance cues and motion models (e.g., Kalman filters) to associate detections, as exemplified by SORT [11], a lightweight tracker that uses bounding box overlap and constant-velocity motion for data association. An improvement over SORT was DeepSORT [1], which adds a deep CNN to compute appearance embeddings for each detected object, making the association more robust to occlusions and crossings. In recent years, transformer-based trackers have redefined the MOT landscape. These methods integrate detection and tracking into a unified architecture, using attention mechanisms to model object interactions over space and time. For example, TrackFormer [2] treats object tracks as learnable queries and carries them through a transformer decoder from frame to frame. TransTrack [3] similarly introduces a transformer that takes both previous frame tracks and current frame detections as input to predict new positions. MOTR [5] and its improved version MOTRv2 [12] eliminate the separate detection stage (they use a set of track queries in a DETR-like transformer to simultaneously detect new objects and update existing tracks in each frame). Although these end-to-end approaches demonstrate excellent performance on benchmarks, their high complexity requires powerful hardware. Even these advanced trackers struggle with prolonged occlusions or crowded scenes when limited to one camera view [13]. Lightweight association strategies, such as ByteTrack, OC-SORT, and GIAOTracker, improve MOT by strengthening motion-centric matching and fragmentation repair without requiring costly deep components. ByteTrack employs a dual-threshold policy that leverages low-score detections only to continue existing tracks. OC-SORT introduces observation-centric re-updates and a direction-aware cost to handle occlusions and non-linear motion. GIAOTracker, on the other hand, combines online association, a global link stage, and post-processing informed by camera/object motion to reconnect broken tracklets.

B. MULTI-CAMERA TRACKING

Multicamera multi-object tracking (MCT) introduces an inter-camera data association problem in addition to individual camera tracking. A common strategy, reflected in the AI City Challenge and other benchmarks, is a distributed pipeline: perform an independent MOT on each camera, and then match the resulting trajectories across cameras. The matching step can be formulated as a graph problem or clustering problem. For instance, TRACTA [14] utilises a tracklet affinity measured by both appearance and temporal

proximity, and solves cross-camera matching via graph clustering with robust non-negative matrix factorisation. ReST performs two-stage association (coarse spatial clustering across views followed by temporal refinement), mitigating single-view ambiguities. Typically, appearance features (ReID embeddings) are a key input to these algorithms, often combined with spatial-temporal constraints (e.g. cameras that are far apart in distance or view likely have longer time gaps for the same vehicle to travel between, so a matching must respect plausible travel time) [7]. Reference [15] explicitly addresses the challenges of low-resolution CCTV and computational constraints on edge devices. Their method, ranked among the top five in the 2021 NVIDIA AI City Challenge, achieves large-scale city-wide tracking by combining efficient detection, ReID, and tracking modules optimised for real-world latency and resource limitations. This work demonstrates that high accuracy can be retained without relying on powerful hardware or high-resolution video, although it needs a GPU for multi-object tracking.

Another line of work in multicamera tracking leverages additional contextual cues. One example is trajectory prediction for association, which uses a future trajectory predictor to estimate where a person (in their case) might reappear and integrate that into the matching metric for cross-camera tracking. This is particularly helpful in large camera networks, where a target might temporarily disappear; a motion model can narrow the search space of possible reidentities [16]. In traffic surveillance, an analogous idea is to use the topology of the road network. Reference [17] presents a system that executes deep learning detection and tracking algorithms directly on camera nodes and communicates vehicle metadata via low-power short-range radio, enabling decentralised, privacy-preserving tracking suitable for parking lots and intersections. However, it does not utilise appearance-based ReID, which makes identity association weak when cameras are spatially disjoint or the angles differ significantly from deployment to deployment. License Plate Recognition (LPR) is another powerful cue in vehicle tracking that has been used in practical systems: reading the plate gives a direct ID, which can significantly simplify multicamera matching (vehicles with the same plate are considered identical). However, LPR is not infallible (plates can be obscured or unreadable in motion), and it must be combined with visual matching for cases where the plate is missing or duplicated.

Bird's Eye View (BEV) and V2X-based Tracking: An emerging trend, especially in autonomous driving research, is to combine multiple camera views into a unified bird's-eye view representation for detection and tracking. For example, BEVFormer [18] and PETR [19] employ transformers to aggregate features from multiple cameras and time steps to detect all objects in a global BEV coordinate frame. This approach can naturally handle multicamera tracking in the sense that, once objects are located in BEV with a unique identifier, switching cameras is just observing the same BEV track from different angles. However, the

deployment of the current proposal does not benefit from the commodity of such BEV projections. Unlike automotive platforms equipped with calibrated surround-view cameras, the Mercabarna CCTV network consists of independently mounted cameras positioned in static, asymmetric locations (often at the corners of buildings, with no overlapping fields of view or known extrinsic calibration).

C. Reidentification MODELS

Reidentification (ReID) aims to match the same object across different cameras or viewpoints by learning robust embeddings that preserve identity despite variations in angle, illumination, and other factors. Recent progress [20] has focused on the adoption of Vision Transformers (ViT) [21]. In particular, TransReID [22] achieved state-of-the-art results on vehicle datasets such as VeRi-776 and VehicleID [23] by enhancing a ViT-base backbone with ReID-specific modules.

A key characteristic of modern ReID systems is the use of multi-stage training strategies to maximise feature discrimination and improve cosine similarity between embeddings of the same object. Typically, this involves three steps: (1) initialising with a robust pre-trained backbone (e.g., ViT trained on ImageNet [24] or large-scale self-supervised data), (2) fine-tuning on an intermediate supervised dataset related to the target domain, and (3) performing final fine-tuning on the target ReID dataset with metric-learning objectives such as triplet or contrastive loss.

Several works illustrate the benefits of this progressive approach. For example, [25] introduced a multi-domain learning pipeline in which a ResNet-based model was first trained on a mixture of real and synthetic vehicle images, then fine-tuned on real-world data while freezing lower layers. Similarly, [26] demonstrated that incorporating diverse synthetic vehicles during training yields more discriminative embeddings and stronger generalisation in real-world deployment scenarios. VehicleNet [27] further exemplifies this principle by applying a two-stage progressive scheme: first training with a classification loss on a large mixed-source dataset to learn generic representations, then fine-tuning on the target domain to refine embeddings. Within the ViT paradigm, self-supervised pretraining has also shown substantial benefits. For instance, [28] proposed pretraining a ViT model on the large-scale unlabeled dataset using self-supervised objectives, followed by supervised fine-tuning on ReID datasets. This multi-stage strategy effectively narrows the domain gap and improves downstream embedding performance. More recently, [29] leveraged a pretrained ViT and applied multi-aspect-ratio fine-tuning with feature fusion, showing that specialised intermediate training can enhance robustness in vehicle ReID tasks. These results highlight the importance of staged fine-tuning and domain adaptation in achieving robust ReID performance.

Beyond ViT-based pipelines and staged fine-tuning, recent works have focused on strengthening discriminative cues and evaluation protocols for vehicle ReID under challenging

TABLE 2. Comparison of multi-camera vehicle tracking (MCVT) approaches across deployment assumptions, topology requirements, ReID strategies, and evaluation paradigms.

Work	CPU-based MOTR	Overlapping Field of View (FOV)	Edge/Fog Suitability	ReID Strategy	Evaluation
[7]	No	Not required	Yes	Deep ReID Spatio-temporal constraints	Benchmark-driven
[10]	No	Beneficial	No	Deep ReID	Benchmark-driven
[13]	No	Beneficial	No	Deep ReID Spatial-temporal constraints Graph constraints	Benchmark-driven
[14]	No	Not required	No	Deep ReID Graph constraints	Benchmark-driven
[15]	Yes	Not required	Yes	Deep ReID Spatio-temporal constraints	Deployment-aware
[16]	No	Required	No	Trajectory prediction	Benchmark-driven
[17]	Yes	Not required	Yes	Spatio-temporal constraints	Deployment-driven
[18]	No	Required	No	BEV Graph constraints	Benchmark-driven
[19]	No	Required	No	BEV Graph constraints	Benchmark-driven
Our Work	Yes	Beneficial	Yes	Deep ReID Trajectory prediction Spatio-temporal constraints Graph constraints	Deployment-driven

CCTV conditions. MDFE-Net [30] which enhances multiple discriminative features via a non-local attention block and augments training with centre and weighted regularisation terms. They also propose a complementary metric, mean Positive Sample Occupancy (mPSO), to better characterise retrieval behaviour. MDFE-Net reports strong results on standard vehicle ReID benchmarks, including VeRi-776, VRIC, and VehicleID, underscoring its robustness across scene variability common to city-scale deployments. Complementary to model design, image quality remains a limiting factor in real-world networks such as VRIC [9], where motion blur and low resolution hinder appearance matching. While not a ReID model per se, the Multi-Branch Feature Fusion Super-Resolution network (MBFSR) [31] exemplifies modern SISR that can serve as a pre-processing stage to recover fine-grained details (e.g., lights, badges, and subtle textures) before embedding extraction. Integrating super-resolution selectively can improve downstream ReID stability without incurring prohibitive latency.

III. OUR APPROACH

This section details the architecture and components of the proposed multi-camera vehicle tracking system. We describe

the modular pipeline designed for edge and fog deployment, covering its three main modules: vehicle detection, feature extraction, and trajectory reconstruction with reidentification.

A. SYSTEM OVERVIEW

The proposed multi-camera vehicle tracking system is designed with a modular and scalable architecture that emphasises edge computing for enhanced efficiency. Fig. 1 provides an overview of the pipeline, which consists of three main modules: (1) Vehicle Detection, (2) Vehicle Feature Extraction and (3) Trajectory Reconstruction & Reidentification. Each module operates semi-independently, allowing upgrades or replacements without affecting the others. Specifically, a nearby edge computing unit that runs the detection (one edge computing unit can run one or more camera feeds) processes each camera feed, while a lightweight central (fog) node can extract features of specific vehicles or aggregate metadata for cross-camera tracking. The modular edge-based approach also ensures scalability: additional cameras can be added simply by deploying new edge modules, with the fog node handling the increased flow of vehicle metadata in a distributed fashion. Importantly,

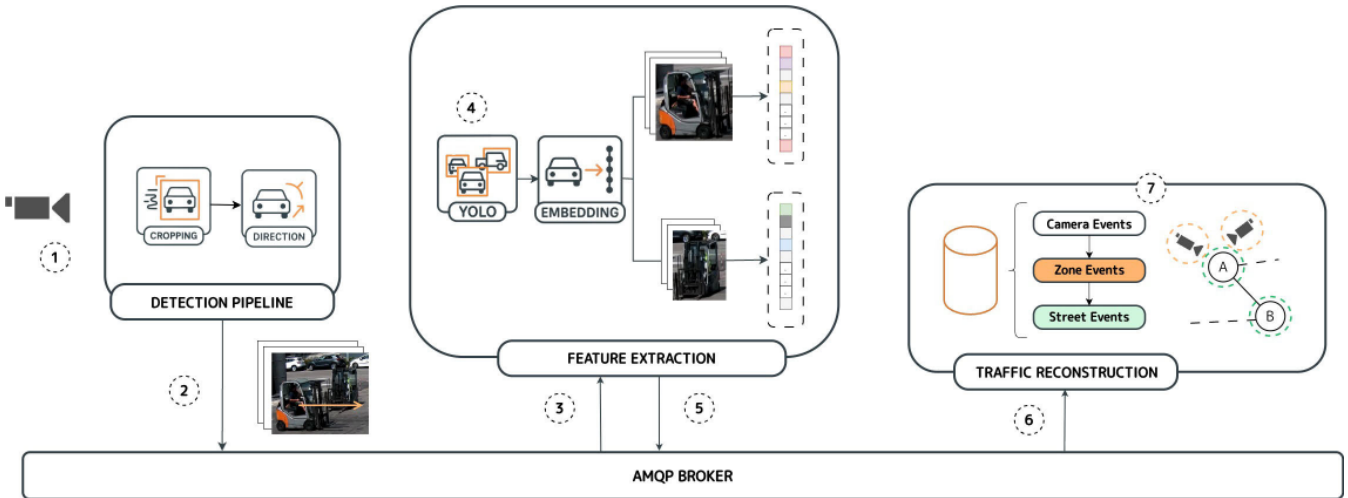


FIGURE 1. System overview of the proposed multi-camera vehicle tracking pipeline. The architecture consists of three main modules deployed over an edge-fog infrastructure. (1) Each edge camera stream is processed by the detection pipeline, which performs cropping and direction estimation to extract moving vehicles. (2) Detected vehicle crops are sent via the AMQP broker to the feature extraction module, where (3) metadata and images are received asynchronously. (4) A YOLO-based detector filters false positives and classifies vehicle types, followed by a vision transformer embedding model that generates compact feature vectors for reidentification. (5) The resulting embeddings and directional metadata are sent to the traffic reconstruction module (6), which aggregates them into camera events. The system periodically updates these events to link cameras (forming zone events) and subsequently match zones to reconstruct complete vehicle trajectories across the monitored area (street events).

computation is performed close to the data source (cameras), in line with the edge computing paradigm, which improves real-time responsiveness and reliability in the logistics hub environment.

B. VEHICLE DETECTION PIPELINE

The system reduces bandwidth and latency by executing initial vehicle detection on fog devices using traditional lightweight computer vision methods. These procedures operate efficiently without requiring GPUs, making the system suitable for edge processing environments. The choice of traditional approaches is motivated by several practical considerations: (1) they demand very low computational resources, which makes them well-suited for edge devices with limited hardware capacity or legacy CCTV cameras; (2) they are straightforward to tune without large-scale training, simplifying deployment and adaptation to new camera setups; (3) background subtraction enables low-cost extraction of regions of interest (ROIs) before applying more expensive detection stages, thereby reducing computation when most of the scene remains static; and (4) they optimize memory usage, since only cropped candidate regions need to be stored or processed, minimizing RAM consumption and I/O overhead when disk-based storage is required. The detection pipeline consists of four stages: (1) bounding box identification, (2) postprocessing, (3) real-time tracking, and (4) direction identification.

1) BOUNDING BOX IDENTIFICATION

The pipeline applied to each camera image combines multiple algorithms, as illustrated in Fig. 2, starting with the identification of bounding boxes. To identify bounding boxes, a background subtraction method is employed using

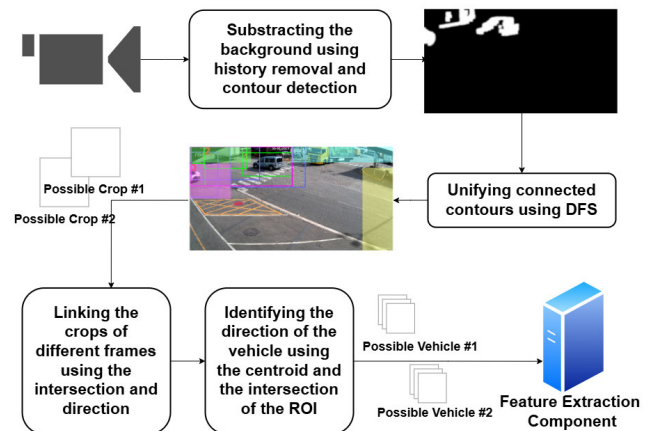


FIGURE 2. End-to-end vehicle detection pipeline on the edge/fog stack. The diagram summarises the four detection stages: bounding-box identification (via background subtraction), post-processing, real-time tracking, and direction identification.

previous frames as references, as the objective is to detect only vehicles. The overall background subtraction process is illustrated in Fig. 3, the algorithm models each pixel in the frame as a composition of various Gaussian distributions, which represent the different colours that appear in that pixel over time. The goal is to differentiate between the pixel colours that belong to the background and those that do not. Then, we apply a threshold for each pixel to determine whether it is part of the background distribution. At this stage, the result is a dot mask. To aggregate these dots and connect contours, initial smoothing with a 3×3 kernel is performed three times to minimise mask noise. Subsequently, an iterative method is applied for dilating the mask, accentuating object areas. After dilating the dots, a threshold for values above 100 is imposed. This step is

crucial to enhance the mask's clarity and to isolate some background elements that might be mistaken for the objects being detected. Once parts of the vehicles are identified, they undergo further dilation and smoothing, culminating in the unification of these segments, thereby distinguishing the object from the background.

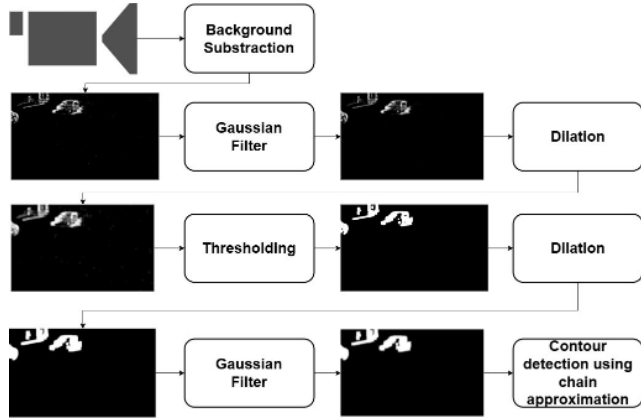


FIGURE 3. From motion mask to candidate vehicles. Example showing background subtraction and morphological filtering to form contours.

2) POST PROCESSING

Once the algorithm identifies the contours, they are passed to bounding rectangles to determine if there are contours that can be grouped. This step is necessary because background subtraction frequently produces fragmented blobs for a single physical vehicle, especially in low-resolution CCTV or under weak texture conditions. For example, large trucks or vans with uniform surfaces (e.g., white trailers) may generate motion response mainly on their borders or shadow edges, while interior regions remain partially absorbed by the background model. As a result, a single vehicle can appear as multiple small bounding boxes that should not be treated independently. If these fragments are not merged, they can lead to duplicated track initializations, unstable crop geometry, and increased identity fragmentation in downstream tracking and reidentification stages.

The postprocessing stage examines those bounding rectangles that meet a specific percentage of intersection, grouping spatially consistent fragments into a single candidate vehicle region. Since grouping should be transitive (i.e., if contour *A* overlaps with *B*, and *B* overlaps with *C*, all three should belong to the same vehicle), the system applies depth-first search (DFS) as a clustering technique, representing contours as nodes and intersections between those contours as edges. Finally, the graph is divided into the different connected components explored, with each one of them forming a subgraph that encompasses the potential vehicle bounding box.

3) REAL-TIME TRACKING

For each frame, the algorithm evaluates every detected bounding box for potential associations with existing vehicle tracks. The SORT algorithm is elegant and fast, but it

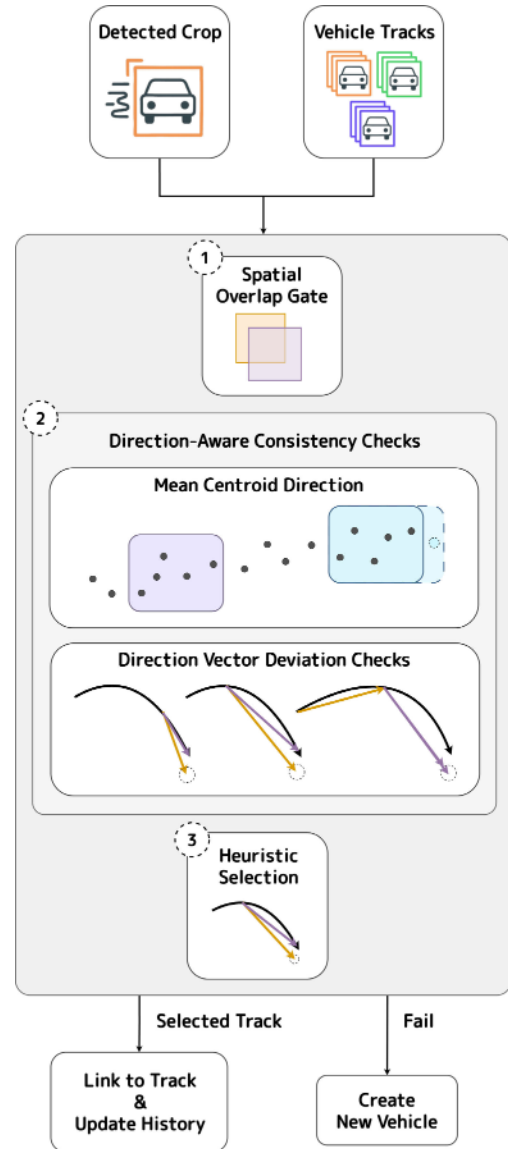


FIGURE 4. Direction-aware centroid linker for real-time tracking. For each detected crop, candidate associations with existing vehicle tracks are evaluated through three stages: (1) a spatial overlap gate based on bounding-box intersection; (2) direction-aware consistency checks using aggregated centroid groups and multi-scale direction vectors; and (3) a heuristic selection step that assigns the crop to the track that minimises directional deviation, or initialises a new track when no valid association exists.

has its drawbacks, especially in complex tracking scenarios. Limitations such as long-term association, short track lifetimes, and abrupt velocity changes are precisely what motivated the creation of an algorithm to track the bounding boxes detected. To address these conditions in an edge-friendly manner, we employ a lightweight *direction-aware centroid linker* that combines (1) a spatial overlap gate, (2) directional consistency checks at different temporal horizons, and (3) a simple angular heuristic to decide between competing matches. The overall association logic of the proposed direction-aware centroid linker is illustrated in Fig. 4.

Let b_t denote a bounding box detected at time t , and let b_{t-1}^v represent the most recent bounding box associated with vehicle track v . A detected crop is considered eligible for association with track v only if a minimum spatial overlap criterion is satisfied, as defined in (1):

$$\text{IoU}(b_t, b_{t-1}^v) > \tau_{\text{IoU}}, \quad (1)$$

where τ_{IoU} is a fixed threshold. To obtain stable motion estimates, directions are not computed from individual centroid pairs but from groups of centroids aggregated over different portions of the track history. Specifically, given a selected history segment, two representative mean centroids are computed by averaging a group of centroids at the beginning and at the end of the segment. A direction vector (2) is then defined as the normalised displacement between these two mean centroids:

$$\mathbf{d} = \frac{\bar{\mathbf{p}}_{\text{end}} - \bar{\mathbf{p}}_{\text{start}}}{\|\bar{\mathbf{p}}_{\text{end}} - \bar{\mathbf{p}}_{\text{start}}\|}, \quad (2)$$

where $\bar{\mathbf{p}}_{\text{start}}$ and $\bar{\mathbf{p}}_{\text{end}}$ denote the mean centroids at the beginning and at the end of the history segment, respectively. This aggregation reduces sensitivity to local jitter and noise while preserving the dominant motion trend of the vehicle. Applying the same procedure to different portions of the centroid history yields short-, medium-, and long-term direction estimates, which capture motion consistency at different temporal scales. These direction estimates are used to assess whether a vehicle follows a coherent path both locally and globally, which is particularly important in intersection areas where multiple trajectories may spatially overlap. In this context, consistency at short distances ensures physically plausible frame-to-frame motion, while consistency at longer distances enforces global trajectory coherence. For each candidate crop, an additional direction estimate, denoted $\mathbf{d}_{\text{new}}$, is computed by hypothetically appending the crop centroid to the track history and recomputing the aggregated direction over the corresponding history segment. Unlike the existing direction estimates, $\mathbf{d}_{\text{new}}$ explicitly encodes the directional change introduced by the candidate association. The effective history length used for this appended estimate is adapted to the track speed, with longer histories for slower motion and shorter histories for faster motion, balancing stability and responsiveness. Directional compatibility between two direction estimates is evaluated through their angular deviation, defined as in (3):

$$\Delta\theta(\mathbf{d}_a, \mathbf{d}_b) = \arccos(\mathbf{d}_a^\top \mathbf{d}_b), \quad (3)$$

where $\mathbf{d}_a$ and $\mathbf{d}_b$ are unit direction vectors derived from aggregated centroid groups. For convenience, the corresponding direction angles are denoted by $\theta_{\text{mid, last}}^v(t)$ for the medium-term direction of track v before association, and by $\theta_{\text{mid, new}}^v(t; b_t)$ for the medium-term direction obtained after hypothetically appending the candidate crop b_t . These angles are directly derived from the corresponding direction vectors via their orientation. This angular deviation is used to perform three complementary consistency checks. First,

a *short-term change check* compares the current short-term direction of a track with the corresponding $\mathbf{d}_{\text{new}}$, allowing the detection of abrupt, physically implausible direction changes. Second, a *medium-term stability check* performs the same comparison at a longer temporal scale, preventing gradual drift or identity switches once sufficient trajectory history has been accumulated. Third, a *long-term consistency check* compares the direction estimated from the initial portion of the trajectory with direction estimates obtained from later portions after appending the candidate crop, enforcing global coherence of the track over extended time spans. A candidate association is rejected when one or more of these angular deviations exceed predefined thresholds. Together, these checks allow tolerance to small local fluctuations while suppressing associations that would contradict the established motion of the vehicle, including cases where different vehicles share a similar spatial path but move in opposite directions.

Among the remaining admissible candidates, association is decided using a simple angular heuristic derived from the medium-term motion of each track. The heuristic score is defined as the wrapped absolute angular difference between the medium-term direction of the track before association and the medium-term direction obtained after hypothetically appending the candidate crop, as defined in (4):

$$h(v, b_t) = |\text{wrap}_{180}(\theta_{\text{mid, new}}^v(t; b_t) - \theta_{\text{mid, last}}^v(t))|. \quad (4)$$

Here, $\text{wrap}_{180}(\cdot)$ maps angular differences to the interval $[-180^\circ, 180^\circ]$ to account for angular periodicity. Smaller values of $h(v, b_t)$ indicate stronger directional consistency between the candidate crop and the existing trajectory. For each detected crop, the vehicle track that minimises the heuristic score is selected as the best match:

$$v^* = \arg \min_v h(v, b_t). \quad (5)$$

If no existing track satisfies the spatial overlap and directional consistency constraints, a new vehicle track is initialised. Tracks are retained for a limited number of frames even in the absence of new associations, improving robustness to short-term occlusions and intermittent missed detections while keeping the overall computation lightweight for real-time edge deployment.

The algorithm first verifies if there is sufficient spatial overlap between the new crop and the previous bounding boxes. For each existing vehicle with spatial overlap, a short, medium and long speed-adaptive temporal direction of its movement is maintained, allowing the system to measure how much the direction would change if the new crop were added to the track. A crop is rejected if: a) it causes a significant short-term direction and either a significant long-term change or middle-term change; or b) it causes a significant middle-term direction change when the trajectory is long enough. The multi-scale approach enables maintaining object identity over extended periods, even in complex scenarios with multiple objects and varying motion patterns.

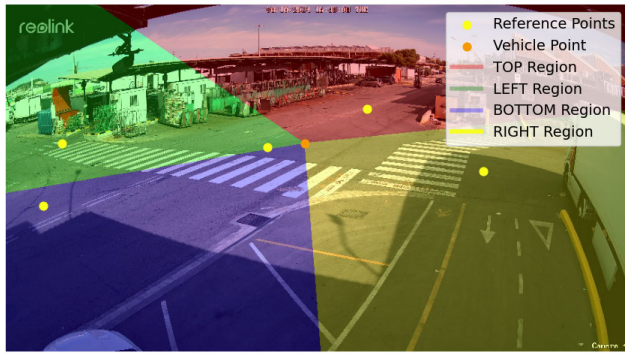


FIGURE 5. Direction labelling by weighted interpolation. Given a position p , distances and importance weights to reference points define an interpolated angular range that maps to the most likely motion label (UP/RIGHT/DOWN/LEFT).

4) DIRECTION IDENTIFICATION

Once the system has linked all bounding boxes corresponding to a single vehicle, it calculates the vehicle's trajectory orientation, identifying the start and end directions (e.g. UP, LEFT, RIGHT or DOWN). First, two angles are computed: one from the first centroid to the midpoint (start angle) and another from the midpoint to the final centroid (end angle). Since each camera may have a different orientation relative to the global scene, the measured angles are transformed using a camera-specific directional mapping to ensure consistency in labelling across views. This is achieved by defining a set of reference points, each associated with a predefined angular range representing a directional label, as illustrated in Fig. 5. The system computes a weighted combination of all reference points. Thus, for a detected position, the algorithm evaluates its relationship to each reference point using a weight w_i as defined in (6):

$$w_i = \left(\frac{1}{\text{dist}(p, c_i) + \varepsilon} \right) \cdot \alpha_i, \quad (6)$$

where $\text{dist}(p, c_i)$ is the Euclidean distance between the vehicle's point p and reference point c_i ; ε , a small constant to prevent division by zero; and α_i , a predefined importance factor reflecting the relevance for each point due to the camera's point of view. These weights are subsequently normalized as in (7):

$$w'_i = \frac{w_i}{\sum_{j=1}^n w_j}. \quad (7)$$

Given the known directional range associated with each reference point, defined as an interval $[\text{from}_i(D), \text{to}_i(D)]$, the final estimated directional range for the trajectory is computed as in (8):

$$\text{Range}(D) = \left[\sum_{i=1}^n w'_i \text{from}_i(D), \sum_{i=1}^n w'_i \text{to}_i(D) \right]. \quad (8)$$

Finally, the system determines which directional label best matches the angle of the vehicle's motion vector by checking if it falls within any of the interpolated directional

ranges. This method enables robust and context-aware direction labelling, accommodating camera-specific layouts and complex orientation scenarios.

C. FEATURE EXTRACTION COMPONENT

Once vehicles are detected, the system enriches each detection with additional features for reidentification. This component refines the raw bounding box detections and prepares the data for reidentification tasks. The first step involves applying a YOLO-based object detector to the cropped image (trained using a mixture of public datasets [32], [33], [34], [35], [36]), with a class distribution as described in Fig. 6. This step serves as a filtering and classification stage, removing false positives from the initial detection and classifying valid crops into predefined classes: car, van, truck, forklift, and motorcycle. Additionally, YOLO enables the detection of multiple vehicles within the same crop, a situation that may arise due to limitations in the initial vehicle detection stage. In such cases, a single crop can lead to multiple valid vehicle detections, ensuring that no instance is missed.

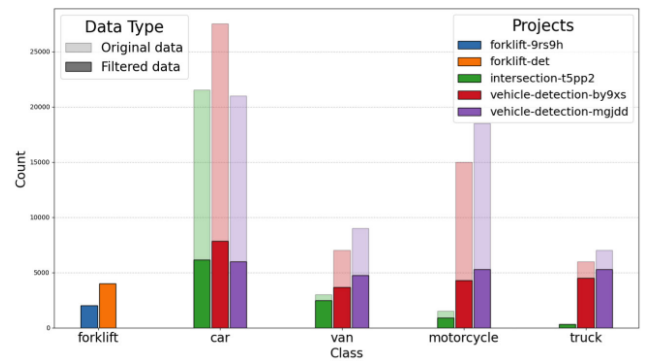


FIGURE 6. Class distribution and balancing for YOLO training. Dataset composition across car, van, truck, forklift, and motorcycle and the balancing strategy used to mitigate class imbalance during detector training.

For each valid crop, the system generates a vehicle embedding using a fine-tuned ViT-B/16 that captures high-level semantic features of vehicle appearance. The backbone used is a ViT-B/16 [37] pre-trained on ImageNet-21k and fine-tuned on ImageNet 2012 [24], both at resolution 224×224 . By our part, the model is fine-tuned using the Stanford Cars dataset [38], [39] and then, to improve its suitability for the reidentification task in Mercabarna's specific logistics context (appearance of new classes such as forklifts, motorcycles and trucks), the model was fine-tuned with a triplet-based contrastive loss on a private dataset using LoRa. This training approach encourages embeddings of the exact vehicle (from different angles or frames) to be close in feature space, while pushing apart embeddings of different vehicles. The LoRa configuration was selected to balance adaptation capacity and overfitting control given the limited size of the Mercabarna dataset. Table 3 and Table 4 show fine-tuning training parameters used. Although the Mercabarna dataset is relatively small, it stands out for

its representativeness in real-world logistics environments, where backgrounds and vehicle types create a domain that is difficult to reproduce. We have reduced the learning rate in this stage to prevent overfitting and promote better model adaptation.

TABLE 3. Fine-tuning training parameters for embedding models. Learning rates, training steps/epochs, and sample counts for the Stanford Cars pre-fine-tuning and the Mercabarna triplet-loss adaptation used to produce the final ViT embedding model.

Dataset	LR	Steps or epochs	Training samples
Stanford Cars	1×10^{-5}	40000 steps	8140 images
Mercabarna	5×10^{-5}	28 epochs	3481 images of 907 vehicles

TABLE 4. LoRA configuration for ViT embedding fine-tuning. Low-Rank adaptation hyperparameters used during the Stanford Cars pre-fine-tuning and subsequent Mercabarna triplet-loss adaptation to produce the final vehicle re-identification embedding model.

Parameter	Value
Rank (r)	8
Alpha (α)	32
LoRA dropout	0.05

The embedding model is used to group the vehicles detected by YOLO across different frames. To associate detections across crops, we compute the cosine similarity between two embedding vectors e_a and e_b as in (9):

$$S_{\cos}(e_a, e_b) = \frac{e_a^\top e_b}{\|e_a\| \|e_b\|}, \quad (9)$$

where $S_{\cos} \in [-1, 1]$ measures appearance similarity. Two crops are considered to represent the same vehicle if $S_{\cos} > \tau_{\text{reid}}$, with τ_{reid} denoting the reidentification threshold, allowing the system to build consistent tracklets over time. Note that this association can only be performed across YOLO-derived crops from different frames, and not within the same traditional crop where multiple vehicles may appear simultaneously. Lastly, the feature extraction pipeline concludes by computing and storing the average embedding across the tracked trajectory for matching vehicles across disjoint camera views.

D. TRAJECTORY RECONSTRUCTION

This module, responsible for reconstructing vehicle trajectories across different camera views, uses the embeddings and metadata (camera ID, timestamp, and directions) produced in the previous stages to establish spatio-temporal links between detections. The process begins by matching vehicle detections from different cameras that are part of the same predefined zone. These matches are called *zone events*. In cases where cameras have overlapping views, the same vehicle may be detected simultaneously by two cameras. The system identifies these duplicate instances

using a combination of high embedding cosine similarity and near-synchronised timing. Such duplicates are resolved by consolidating them into a single track so that the vehicle is counted only once. In case of ambiguous directions, both directions are discarded and the direction of the new event is set to unknown. If no match is found between two cameras in the same zone within a specified temporal range, the unmatched events are still stored for further analysis, but with lower priority. Vehicles in zones with only one camera are always stored as zone events.

Once candidate matches are stored, they serve as the foundation for re-identification across zones and the reconstruction of complete trajectories. A link between zone events is called a *street event*, and a trajectory is defined as a set of linked street events. Mercabarna's camera layout provides natural constraints: cameras cover partially overlapping or sequential zones such that a vehicle travelling through the hub will appear in a predictable order. These spatial relationships are encoded as prior knowledge. Thus, when a new detection occurs, the search for a match is restricted to a pool of candidate embeddings from cameras that could logically observe the same vehicle next. Additionally, temporal constraints are imposed based on the estimated travel time between zones, which reduces false match attempts by pruning physically implausible pairings.

To further control false associations, we use two cosine similarity thresholds depending on the reliability of the motion metadata. When the street directions are known and consistent, directionality acts as an additional discriminative cue that reduces the effective ambiguity of the candidate set. Therefore, a more permissive appearance threshold is sufficient (e.g., $\tau_{\text{dir}} = 0.65$), allowing matches under challenging viewpoint/illumination changes while remaining constrained by direction feasibility. In contrast, when direction is unknown (e.g., due to reversing, stops, or conflicting motion estimates), the system must rely almost exclusively on appearance similarity. In this regime, we adopt a stricter threshold (e.g., $\tau_{\text{unk}} = 0.85$) to compensate for the increased uncertainty and to avoid visually plausible but incorrect cross-camera merges, which are common under low-resolution CCTV and repetitive vehicle fleets.

IV. EXPERIMENTS AND RESULTS

This section evaluates the YOLO refinement detector on a Mercabarna validation set, then analyzes embedding stability and cross-dataset retrieval performance on VeRi-776 and VRIC. Next, we report end-to-end multi-camera identity metrics on Mercabarna using manually tagged matches, followed by a targeted ablation study isolating tracking, refinement, and embedding contributions. Finally, we describe the full-scale deployment across 27 cameras and report runtime resource requirements.

A. YOLO DETECTION MODEL

To evaluate the detection prior to the feature embedding stage, we created a validation dataset from the Mercabarna deploy-

TABLE 5. Performance of the YOLO-based vehicle detector on the Mercabarna validation set, showing per-class recall and mAP@0.5 across 22474 instances, highlighting robust forklift/truck recall and lower recall on motorcycles due to size/occlusion.

Class	Instances	Recall	mAP@0.5
Forklift	7743	0.862	0.904
Car	4834	0.745	0.843
Truck	4342	0.821	0.816
Van	3432	0.673	0.723
Motorcycle	2123	0.593	0.691
All	22474	0.739	0.791

ment. The validation consisted of 8964 images containing instances with the trained class distributions. Images of parked vehicles were removed to assess only vehicles in movement. Additionally, 17% of the images in the validation set do not have any labelled instances.

Table 5 summarizes the per-class recall and mean Average Precision (mAP@0.5) on the Mercabarna validation set. Fig. 7 complements these results by showing the main inter-class confusions under row-normalization, independent of class imbalance. As shown in Table 5, despite the class imbalance, the model performed reliably across all categories. We used the mAP as the main evaluation metric, since it provides a balanced measure that captures both precision and recall across classes and IoU thresholding. Forklifts, which are particularly relevant in warehouse operations, achieved the highest mAP@0.5. Meanwhile, motorcycle detection showed the lowest recall, highlighting the difficulty of detecting smaller or more dynamic objects in occluded scenes. As shown in Fig. 7, the row-normalized confusion matrix further reveals the main misclassification patterns, with motorcycles being more frequently confused with larger vehicle categories, which is consistent with their lower recall. Overall, the model achieved strong detection performance across most classes, with forklifts and trucks obtaining the highest recall rates.

B. VEHICLE EMBEDDINGS

To evaluate the effectiveness of the vehicle feature extraction strategy for cross-camera vehicle reidentification, we compared the three ViT models by analysing the distribution of cosine similarities between embedding vectors of the same vehicle across frames. The validation dataset consisted of 325 images of 76 vehicles. The baseline model shows a broad and dispersed distribution, indicating poor consistency in embeddings. Fig. 8, illustrates the difference in cosine similarity between the three models, highlighting how our Mercabarna fine-tuned model achieves better intra-class closeness and inter-class separation. The Stanford model significantly improves upon this, producing a more concentrated distribution centred above 0.75. The Mercabarna model further tightens this distribution and shifts slightly toward higher similarity values, confirming a marginal but consistent improvement. This performance improvement is

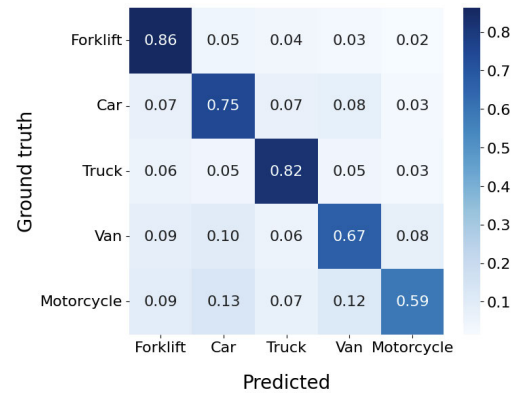


FIGURE 7. Row-normalized confusion matrix of the YOLO-based vehicle detector evaluated on the Mercabarna validation set. Rows correspond to ground-truth classes and columns to predicted classes, with values normalized per row. This representation highlights systematic inter-class confusions independently of class imbalance. In particular, motorcycles exhibit higher confusion with larger vehicle categories, consistent with their lower recall and mAP@0.5 reported in Table 5.

partly attributed to the inclusion of vehicle categories (such as forklifts, vans, and trucks) that are absent in the Stanford Cars dataset. However, increasing the number of epochs, the learning rate, and the number of training iterations could allow for deeper convergence on the domain-specific distribution. Additionally, expanding the validation dataset may help to identify better overfitting or assist in avoiding it during training.

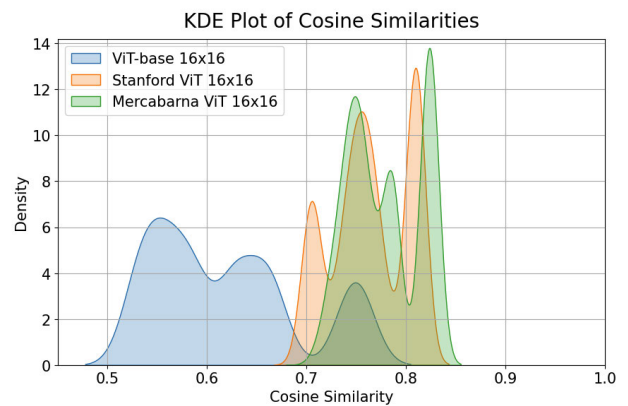


FIGURE 8. Embedding consistency across models. KDE of cosine similarities for ground-truth pairings of baseline ViT-base 16×16 (blue), the Stanford ViT 16×16 fine-tuned on the Stanford Cars dataset (orange), and the Mercabarna ViT 16×16 model (green) in validation dataset.

To assess transferability beyond Mercabarna at the embedding level, we evaluated the appearance model on VeRi-776 [8] and VRIC [9] benchmark dataset. VeRi-776 and VRIC are used here to assess the appearance embedding component in isolation, rather than the full end-to-end Mercabarna system. These benchmarks provide cross-camera vehicle retrieval metrics, whereas our deployed pipeline additionally relies on deployment-specific elements. Consequently, the results below should be interpreted as a measurement of

how well the learned embedding space generalizes to unseen camera networks and vehicle identities, and as evidence that the Mercabarna-adapted representation is not merely memorizing the deployment data. All reported benchmark results are obtained in a zero-shot setting (no fine-tuning on the benchmark training splits).

For VeRi-776, we adopt an evaluation setting in which any image of the same vehicle captured by a different camera is considered a valid match, ignoring timestamp and spatial resolution constraints, as the objective is to validate the robustness of the learned embedding space. We report Precision at various top- k retrieval thresholds and mean Average Precision (mAP). For each query q , let $\mathcal{G}_q$ denote the set of ground-truth matches and let $r_k \in \{0, 1\}$ indicate whether the retrieved sample at rank k is correct. Precision at rank k is defined as (10):

$$\text{Precision}@k = \frac{1}{k} \sum_{i=1}^k r_i. \quad (10)$$

Average Precision (AP) for a single query is computed as in (11):

$$\text{AP}(q) = \frac{1}{|\mathcal{G}_q|} \sum_{k=1}^N \text{Precision}@k \cdot r_k, \quad (11)$$

where N is the number of retrieved gallery samples. The mean Average Precision (mAP) over all Q queries can be found in (12).

$$\text{mAP} = \frac{1}{Q} \sum_{q=1}^Q \text{AP}(q). \quad (12)$$

The zero-shot evaluation on VeRi-776 [8], as it can be found in Table 6, demonstrates strong appearance robustness of the learned embedding space under cross-camera matching conditions. The model achieves very high precision at low retrieval depths, indicating that for the vast majority of queries, the top-ranked retrieval corresponds to the correct vehicle identity. This reflects a well-structured embedding space in which intra-class samples cluster tightly despite viewpoint and illumination variations. The mAP values further confirm that correct matches are consistently ranked within the highest retrieval positions. The gradual decrease in mAP as the retrieval depth increases is expected, as larger retrieval pools introduce more visually similar distractors. Nevertheless, the relatively slow decay suggests that the model maintains strong discriminative capacity beyond the first position and preserves meaningful neighborhood ordering. Importantly, these results were obtained without fine-tuning on VeRi-776, which indicates that the embedding model generalizes effectively to unseen camera configurations and vehicle instances. Furthermore, incorporating spatial consistency constraints would further improve quantitative performance, but at the cost of reducing the strict zero-assumption nature of this evaluation. Overall, given that VeRi-776 includes significant viewpoint changes and

real-world traffic variability, the high Precision@1 and mAP values suggest that the learned representation captures stable vehicle-specific features rather than overfitting to dataset-specific biases.

To further probe cross-dataset generalization, we also evaluated on the VRIC benchmark [9] without using its training split. In addition to mAP, VRIC reports Rank- k accuracy, defined as in (13):

$$\text{Rank-}k = \frac{1}{Q} \sum_{q=1}^Q \mathbb{I}(\exists i \leq k \text{ s.t. } r_i = 1), \quad (13)$$

where $\mathbb{I}(\cdot)$ is the indicator function. Rank- k measures the probability that at least one correct match appears within the top- k retrieved results.

TABLE 6. VeRi-776 [8] retrieval results (zero-shot) assessing appearance robustness.

Metric	Value
mAP@5	0.9439
mAP@10	0.8379
mAP@15	0.7639
precision@1	0.9874
precision@5	0.8510

TABLE 7. VRIC [9] retrieval results (zero-shot), shows solid cross-dataset transfer with remaining domain shift.

Metric	Ours	Baseline [9]	[31]	[30]
Rank-1	53.60	46.61	79.97	80.75
Rank-5	72.53	65.58	94.09	-
mAP	62.19	47.50	82.75	86.58

As reported in Table 7, our model outperforms the original VRIC baseline [9] across all reported metrics despite being evaluated in a strict zero-shot setting. This indicates that the learned embedding space generalizes beyond the training distribution and retains discriminative structure under unseen surveillance conditions. However, methods explicitly trained and optimized on VRIC, such as those proposed by [30] and [31], achieve substantially stronger retrieval performance.

This performance gap suggests the presence of residual domain shift, likely attributable to dataset-specific visual statistics in VRIC, including tighter bounding boxes, compression artifacts, extreme pose variation, and camera-specific illumination patterns. We hypothesize that incorporating a more diverse, cross-domain training corpus (spanning heterogeneous camera geometries and environmental conditions) would improve invariance to such factors and further narrow the gap to VRIC-trained approaches.

These benchmark results are complemented by the deployment-grounded ablation study in Table 9, which demonstrates that domain alignment of the embedding model is the dominant factor for multi-camera identity preservation under Mercabarna conditions. Together, the controlled benchmark evaluation and the real-world ablation

TABLE 8. Multi-camera re-identification on Mercabarna. Identity Recall (IDR), Identity Precision (IDP), IDF1, and Uniqueness computed on 226 manually tagged matches; high recall with precision/uniqueness indicating residual ambiguity from direction cues and occlusions.

Metric	Value (%)
ID Recall (IDR)	86.72
ID Precision (IDP)	78.40
IDF1	82.35
Uniqueness	75.53

analysis consistently indicate that embedding robustness is primarily limited by dataset-specific domain characteristics rather than architectural constraints.

C. REIDENTIFICATION EVALUATION METRICS

To evaluate the whole system under real multi-camera operational conditions in Mercabarna, we computed identity-based metrics using 226 manually annotated vehicle matches as ground truth. Let TP denote correctly predicted identity associations, FP incorrect predicted associations, and FN missed ground-truth associations. Identity Recall (IDR), measuring how many ground-truth re-identification connections are successfully recovered, is defined as (14):

$$\text{IDR} = \frac{TP}{TP + FN}. \quad (14)$$

Identity Precision (IDP), measuring how many predicted matches are correct, is defined as (15):

$$\text{IDP} = \frac{TP}{TP + FP}. \quad (15)$$

The identity F1-score (IDF1), which balances recall and precision and penalizes both false matches and missed links, is defined as (16):

$$\text{IDF1} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} = \frac{2 \cdot \text{IDP} \cdot \text{IDR}}{\text{IDP} + \text{IDR}}. \quad (16)$$

Finally, we define Uniqueness to measure identity fragmentation, as can be defined in (17). Let GT be the number of ground-truth identity trajectories and S the number of predicted identity fragments corresponding to those trajectories:

$$\text{Uniqueness} = \frac{GT}{S}, \quad (17)$$

where a lower value indicates trajectory splitting or duplicated identities.

As it can be seen in Table 8, while our system achieves high recall, the precision and uniqueness metrics reveal limitations in the current reidentification pipeline. The relatively low precision indicates that a considerable portion of predicted matches are incorrect, often resulting from ambiguous embeddings or inaccurate direction labels. IDF1 also indicates a balanced, yet imperfect association across cameras, penalising both missed links and false matches. In highly dynamic environments, such as Mercabarna, forklifts and delivery vehicles frequently exhibit non-linear and bidirectional motion, including reversing or making sharp

turns, which compromises the reliability of direction-based cues. Additionally, temporary stops, unexpected parking, or interactions at pedestrian crossings (e.g., vehicles halting at zebra crossings) can further obscure the actual motion vector. These factors introduce uncertainty in the linking phase and increase the chance of mismatched identities. The uniqueness score, although relatively high, still reflects a notable number of duplicated matches, where a single ground-truth vehicle trajectory is incorrectly split into multiple predicted identities. This fragmentation is often caused by incomplete tracking in low-visibility segments, embedding drift, or trajectory splits due to occlusion or sudden movements. Moreover, inconsistencies in vehicle detection (e.g., false negatives during occluded frames or misclassified static vehicles) can break the temporal coherence needed to group detections reliably. These findings suggest that improving the robustness of the detection and tracking modules (particularly under occlusion and direction ambiguity) is essential for enhancing both precision and identity consistency in future deployments. To improve precision and uniqueness, we propose following recent approaches such as YOLO11-JDE [40], which integrates detection and Re-ID into a single lightweight model, reducing ID switches and enhancing real-time association performance. In edge environments, lightweight models such as YOLOv4-Tiny [41] or ByteTrack [6] combined with optimised tracking strategies (IoU, histograms) have also demonstrated significant improvements in speed and identity uniqueness.

D. ABLATION STUDY

The previous subsections evaluated the main components of the pipeline in isolation and then reported end-to-end multi-camera re-identification performance under real operational conditions in Mercabarna. While these results validate the overall feasibility of the proposed modular architecture, they do not isolate the contribution of key design choices that affect identity preservation across cameras. To this end, we conduct a targeted ablation study focusing on three high-impact components: (i) the single-camera association mechanism responsible for forming temporally stable tracklets that later feed the cross-camera linking stage, (ii) the appearance embedding model used for cross-camera association, and (iii) the YOLO refinement stage that filters and disambiguates detections prior to embedding extraction.

All ablations are evaluated on the same Mercabarna multi-camera protocol described in Section IV, using the same manually annotated set of 226 ground-truth vehicle matches. We report Identity Precision (IDP), Identity Recall (IDR), IDF1, and Uniqueness. Unless explicitly stated, all hyperparameters, camera topology constraints, cross-camera similarity thresholds, and aggregation strategy (track-level average embedding) are held constant to ensure that performance differences can be attributed to the ablated component only.

TABLE 9. Ablation results on Mercabarna multi-camera evaluation. End-to-end identity metrics computed on 226 manually tagged cross-camera matches. Higher is better for IDP, IDR, IDF1, and uniqueness. Values are reported as percentages.

Group	Variant	IDP (%)	IDR (%)	IDF1 (%)	Uniqueness (%)
Baseline	-	78.40	86.72	82.35	75.53
Tracking	IoU-only association	69.33	77.13	73.02	59.63
	SORT tracker	72.90	85.61	78.75	69.54
Refinement	without YOLO-based model	61.24	90.03	72.90	73.86
Embeddings	ViT-B/16	63.48	76.09	69.22	67.26
	Stanford ViT-B/16	75.08	83.09	78.88	74.59
Trajectory Reconstruction	No spatio-temporal constraints	57.50	75.08	65.12	71.29

Table 9 shows that single-camera tracking influences end-to-end re-identification, but it is not the primary determinant of identity precision or recall. Instead, its main impact is on identity fragmentation: simplifying the tracker reduces Uniqueness markedly, indicating that weaker intra-camera association produces shorter and more fragmented tracklets that propagate downstream as duplicated identities. In other words, tracking affects the structural continuity of identities more strongly than the overall ability to recover correct cross-camera links. Disabling the YOLO refinement stage reveals a clear precision–recall trade-off. Removing refinement can slightly increase recall, as more detections are forwarded to the re-identification stage, increasing the chance of recovering ground-truth links. However, this comes at the expense of precision, because the absence of filtering allows false positives and ambiguous crops from the traditional vehicle detection component to enter the embedding and matching pipeline, inflating incorrect associations. These results confirm the role of YOLO refinement as a noise-suppression step that improves the quality of candidate tracklets prior to cross-camera matching. The embedding model is one of the most impactful components for cross-camera identity preservation in terms of both precision and recall. Using a non-fine-tuned ViT embedding leads to a pronounced degradation, demonstrating that domain mismatch yields unstable or insufficiently discriminative representations under Mercabarna’s viewpoint variability and mixed vehicle categories. Fine-tuning on the Stanford Cars dataset improves performance, indicating domain shift due to vehicle types. Comparing it to the baseline, the usage of the Mercabarna fine-tuned model only improves slightly the identity metrics. This is consistent with the KDE of cosine similarities from 8.

In addition, the trajectory reconstruction ablation confirms that the prior topology and travel-time gating encoded in the zone/street event logic is not merely an efficiency heuristic, but a major contributor to association correctness. When spatial ordering constraints (restricting candidate matches to logically reachable cameras/zones) and temporal feasibility constraints (travel-time windows) are removed, the matching problem degenerates into a global appearance-only retrieval over a substantially larger candidate pool. This drastically increases the number of plausible-but-incorrect pairings in Mercabarna, where many vehicles share similar visual signatures (e.g., repeated forklift models, delivery vans, and

trucks). As a result, identity precision collapses and the overall identity consistency degrades, indicating that false cross-camera merges become the dominant failure mode. Interestingly, Uniqueness decreases less sharply than in the weakest tracking ablation, suggesting that the primary impact of removing spatio-temporal constraints is not increased fragmentation, but rather incorrect linking across physically implausible camera transitions. Finally, the recall decrease is consistent with error propagation: early false matches “consume” identities and prevent the correct downstream links from being established, turning what could have been recoverable associations into missed links.

Overall, the ablations indicate that the appearance embedding is the dominant factor shaping multi-camera association quality (IDP/IDR/IDF1), while single-camera tracking mainly affects downstream identity continuity by modulating fragmentation (Uniqueness). The YOLO refinement stage primarily acts as an upstream noise filter that enforces a clear precision–recall trade-off by suppressing false positives and ambiguous crops. Finally, the trajectory reconstruction priors (camera adjacency and travel-time feasibility) prove essential to prevent physically implausible matches, thereby preserving high precision in real deployments, particularly under low-resolution CCTV and visually repetitive vehicle fleets.

E. DEPLOYMENT TEST

We implemented the proposed system in Mercabarna (Barcelona), a major hub of the wholesale market with significant daily truck and vehicle traffic. The deployment consists of twenty-seven CCTV cameras covering key entry/exit points and internal road intersections of the facility, as illustrated in Fig. 9. Seven edge processing units (GEEKOM Mini PC XT12 Pro, Intel i7-12650H, 32 GB RAM) run the Vehicle Detection Pipeline and can be assigned to different cameras depending on stream resolution and activity. The system operates in real time, with edge devices processing each video stream at approximately 20–29 FPS depending on scene activity (high activity corresponds to 6–8 vehicles simultaneously within the ROI). The Mercabarna environment introduces challenging real-world conditions, including strong illumination variation across the day, weather changes, dense traffic, and frequent occlusions due to similar-looking forklifts and delivery vehicles as well as pedestrians crossing the road.

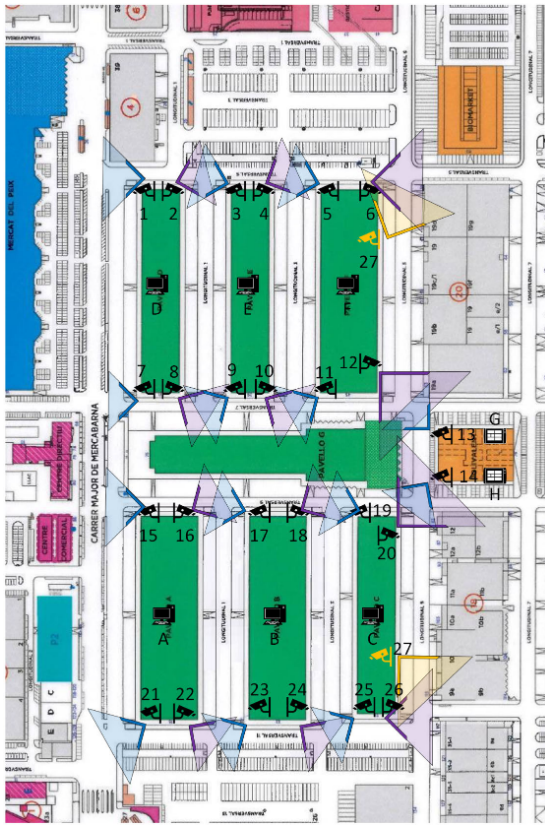


FIGURE 9. Mercabarna multi-camera deployment. Numbered icons mark fixed cameras (1–26) and shaded triangles show their fields of view. LPR checkpoints (27) cover the right-side access corridors. Overlapping views secure cross-street junctions and the central thoroughfare between the main market halls, enabling robust handover for multi-camera vehicle tracking.

On the edge side, the Vehicle Detection Pipeline has a peak RAM usage of 3.65 GB per stream under normal operation and up to 5.65 GB per stream in high-load situations. These peaks occur when vehicles remain visible for long durations, increasing the number of maintained track states and buffered trajectory information; in such cases, a practical trade-off to reduce high-load conditions is to randomly delete (subsample) stored crops per vehicle, bounding the per-vehicle crop buffer while minimally affecting the average embedding representation. After detection, vehicle events are forwarded to the Vehicle Feature Extraction stage, which applies YOLO-based refinement and computes the embedding used for cross-camera association. The feature extraction stage completes one vehicle event (i.e., refinement and embedding for all crops associated with that vehicle) in approximately 1–5 s depending on how long the vehicle is observed and therefore how many crops are processed, typically between 75 and 190 crops per vehicle. With an average inference time of approximately 34 ms per crop (YOLO refinement plus embedding extraction), the effective throughput is about $1/0.034 \approx 29.4$ crops/s, corresponding to an expected per-vehicle processing time of roughly 2.4–5.8 s for 70–170 crops. In practice, the observed 1–5 s

range reflects variability in crop counts across vehicles and streams, as well as load-dependent scheduling of the parallel feature extraction workers.

For the Feature Extraction Component, we instantiated 20 parallel processes on a 40-core Intel(R) Xeon(R) CPU E5-2640 v4 @ 2.40 GHz to sustain the aggregate crop throughput across cameras. A single Trajectory Reconstruction & Reidentification process was sufficient to handle the resulting stream of vehicle embeddings and maintain global trajectories across the full Mercabarna topology in real time.

V. CONCLUSION

This work presented a modular, edge–fog multi-camera vehicle tracking system that decouples detection, feature embedding, and multi-source association to meet real-time constraints under limited compute. Deployed across 27 CCTV cameras with seven edge units sustaining 20–29 FPS, the architecture reduces bandwidth, keeps latency low, and scales by adding nodes per need. In this context, the system provides a practical foundation for deploying city-scale vehicle tracking systems that are efficient, accurate, and resilient, while remaining open to integration with the next generation of detection and association algorithms.

The implementation deliberately prioritises efficiency and operational robustness on resource-constrained edge hardware. However, this design choice introduces a structural limitation. The reliance on traditional computer-vision-based detection and a highly tuned real-time tracking component makes the system closely adapted to the deployment scenario. While this improves stability and throughput, transferring the pipeline to substantially different environments (e.g., altered viewpoints, traffic density, or scene dynamics) may require non-trivial retuning.

From a performance perspective, the ablation analysis revealed a clear component–metric relationship. Appearance representation is the primary driver of multi-camera association quality, directly impacting precision in cross-camera matching. In contrast, the tracking module largely governs identity fragmentation, reflected in the Uniqueness metric. Finally, the detection stage determines the precision–recall balance by controlling false positives and bounding-box consistency. These findings provide a principled roadmap for future improvements.

Building on this foundation, several extensions offer a clear path to improved accuracy and robustness without sacrificing real-time performance. First, future work will focus on migrating the detection stage toward compact, quantized deep neural models. Replacing the current background-subtraction-based approach with lightweight quantized detectors would increase robustness to dynamic urban scenes while preserving real-time performance on the same edge hardware. More stable detections are expected to reduce downstream track fragmentation and improve both precision and Uniqueness without compromising throughput. Second, improving trajectory detection and cross-camera vehicle linking represents a central

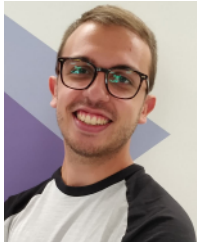
research direction. The ablation results demonstrate that spatial ordering constraints and travel-time feasibility windows are not merely efficiency heuristics but fundamental mechanisms for association correctness. Future extensions will therefore explore more expressive trajectory models that integrate topology priors, dynamic travel-time estimation, and probabilistic gating within a unified formulation. Finally, the re-identification head can be reinforced through more expressive metric-learning objectives and training strategies tailored to viewpoint variation and occlusion. Since appearance representation was identified as the principal driver of multi-camera association quality, improving embedding discriminability through stronger contrastive formulations and domain-adaptive augmentation is expected to yield measurable gains in cross-camera precision.

Overall, the proposed system establishes a practical real-time baseline for edge-deployed MCVT. Its modular architecture enables targeted upgrades at the detector, tracker, or embedding level without architectural redesign. This preserves the efficiency principles of the edge-fog paradigm while providing a controlled and technically grounded path toward higher precision, stronger identity stability, and improved cross-domain transferability.

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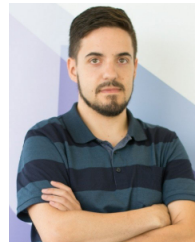
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